

Group no: G_033

Group member: Yuxin Chen, Rishabh Ray

UNF

CATALOGUE (cata_call_no, cata_title, (author_id, author_name), (sub_id, sub_desc),
cata_content, pub_id, pub_name, cate_pub_year, cata_edition, cata_pages,
cata_language, cata_note, (booked_isbn, booked_type))

1NF

CATALOGUE (cata_call_no, cata_title, cata_content, pub_id, pub_name, cate_pub_year,
cata_edition, cata_pages, cata_language, cata_note)

AUTHOR_CATALOGUE(cata_call_no, author_id, author_name)

CATALOGUE_SUBJECT(cata_call_no, sub_id, sub_desc)

BOOKEDITION(booked_isbn, booked_type, cata_call_no)

Candidate Keys:

cata_call_no

booked_isbn

Partial dependencies:

author_id --> author_name

sub_id --> sub_desc

2NF

CATALOGUE (cata_call_no, cata_title, cata_content, pub_id, pub_name, cate_pub_year,
cata_edition, cata_pages, cata_language, cata_note)

AUTHOR_CATALOGUE(cata_call_no, author_id)

CATALOGUE_SUBJECT(cata_call_no, sub_id)

BOOKEDITION(booked_isbn, booked_type, cata_call_no)

AUTHOR(author_id, author_name)

SUBJECT(sub_id, sub_desc)

Transitive dependencies:

pub_id --> pub_name

3NF

CATALOGUE (cata_call_no, cata_title, cata_content, pub_id, cate_pub_year, cata_edition, cata_pages, cata_language, cata_note)

AUTHOR_CATALOGUE(cata_call_no, author_id)

CATALOGUE_SUBJECT(cata_call_no, sub_id)

BOOKEDITION(booked_isbn, booked_type, cata_call_no)

AUTHOR(author_id, author_name)

SUBJECT(sub_id, sub_desc)

PUBLISHER(pub_id, pub_name)

Full dependencies:

cata_call_no --> cata_title, cata_content, cate_pub_year, cata_edition, cata_pages, cata_language, cata_note

booked_isbn --> booked_type, cata_call_no

author_id --> author_name

sub_id --> sub_desc

pub_id --> pub_name

UNF

BORROWER(borrower_no, borrower_fname, borrower_lname, borrower_class,
borrower_phone, borrower_status, (branch_code, book_copy_id, branch_name,
loan_out_datetime, loan_due_date, loan_return_datetime, loan_fine,
loan_fine_paid_date))

1NF

BORROWER(borrower_no, borrower_fname, borrower_lname, borrower_class,
borrower_phone, borrower_status)

LOAN(branch_code, book_copy_id, loan_out_datetime, branch_name, loan_due_date,
loan_return_datetime, loan_fine, loan_fine_paid_date, borrower_no)

CKs:

borrower_no

Partial dependencies:

branch_code --> branch_name

2NF

BORROWER(borrower_no, borrower_fname, borrower_lname, borrower_class,
borrower_phone, borrower_status)

LOAN(branch_code, book_copy_id, loan_out_datetime, loan_due_date,
loan_return_datetime, loan_fine, loan_fine_paid_date, borrower_no)

BRANCH(branch_code, branch_name)

Transitive dependencies: None

3NF

BORROWER(borrower_no, borrower_fname, borrower_lname, borrower_class,
borrower_phone, borrower_status)

LOAN(branch_code, book_copy_id, loan_out_datetime, loan_due_date,
loan_return_datetime, loan_fine, loan_fine_paid_date, borrower_no)

BRANCH(branch_code, branch_name)

Full dependencies:

borrower_no --> borrower_fname, borrower_lname, borrower_class, borrower_phone,
borrower_status

branch_code, book_copy_id, loan_out_datetime --> loan_due_date, loan_return_datetime,
loan_fine, loan_fine_paid_date, borrower_no

branch_code --> branch_name

Collected 3NF

1. CATALOGUE (cata_call_no, cata_title, cata_content, pub_id, cate_pub_year, cata_edition, cata_pages, cata_language, cata_note)
2. AUTHOR_CATALOGUE(cata_call_no, author_id)
3. CATALOGUE_SUBJECT(cata_call_no, sub_id)
4. BOOKEDITION(booked_isbn, booked_type, cata_call_no)
5. AUTHOR(author_id, author_name)
6. SUBJECT(sub_id, sub_desc)
7. PUBLISHER(pub_id, pub_name)
8. BORROWER(borrower_no, borrower_fname, borrower_lname, borrower_class, borrower_phone, borrower_status)
9. LOAN(branch_code, book_copy_id, loan_out_datetime, loan_due_date, loan_return_datetime, loan_fine, loan_fine_paid_date, borrower_no)
10. BRANCH(branch_code, branch_name)

Attribute synthesis

1. CATALOGUE (cata_call_no, cata_title, cata_content, pub_id, cate_pub_year, cata_edition, cata_pages, cata_language, cata_note)
2. AUTHOR_CATALOGUE(cata_call_no, author_id)

3. CATALOGUE_SUBJECT(cata_call_no, sub_id)
4. BOOKEDITION(booked_isbn, booked_type, cata_call_no)
5. AUTHOR(author_id, author_name)
6. SUBJECT(sub_id, sub_desc)
7. PUBLISHER(pub_id, pub_name)
8. BORROWER(borrower_no, borrower_fname, borrower_lname, borrower_phone, borrower_class, borrower_status)
9. LOAN(branch_code, book_copy_id, loan_out_datetime, loan_due_date, loan_return_datetime, loan_fine, loan_fine_paid_date, borrow_no)
10. BRANCH(branch_code, branch_name)

